LEDU_exercise2

Questions

Here is the code provided:

```
pacman::p_load(network, ergm, tnet, install = T)

consult <- Cross.Parker.Consulting.net.info
consult$w <- ifelse(consult$w < 4, 0, 1)
consult <- consult[consult$w == 1, ]
consult <- network(consult[, 1:2], directed = T)
consult %v% "gender" <- Cross.Parker.Consulting.node.gender
consult %v% "region" <- Cross.Parker.Consulting.node.region
consult %v% "orglevel" <- Cross.Parker.Consulting.node.orglevel
```

```
set.seed(510)
ergm.model <- ergm(consult ~ edges + mutual + gwesp(0.5, fixed = T) +
  nodematch("gender") + nodematch("region") +
  nodematch("orglevel") + nodeifactor("orglevel") +
  nodeofactor("orglevel") + absdiff("orglevel"), control = control.ergm(
    MCMC.samplesize = 5000,
    MCMC.burnin = 10000,
    MCMLE.maxit = 10
    ),
    verbose = T)
```

```
Evaluating network in model.

Initializing unconstrained Metropolis-Hastings proposal: 'ergm:MH_SPDyad'.
Initializing model...
Model initialized.
Using initial method 'MPLE'.
```

```
Initial parameters provided by caller: None.
  number of free parameters: 15
  number of fixed parameters: 0
Fitting initial model.
Starting maximum pseudolikelihood estimation (MPLE):
Obtaining the responsible dyads.
Evaluating the predictor and response matrix.
Maximizing the pseudolikelihood.
Finished MPLE.
Starting Monte Carlo maximum likelihood estimation (MCMLE):
Density guard set to 10000 from an initial count of 168 edges.

Iteration 1 of at most 10 with free parameter vector:
                edges                mutual    gwesp.OTP.fixed.0.5
          -3.70825456            3.38833650             0.71406827
     nodematch.gender      nodematch.region     nodematch.orglevel
           0.04650997            0.03850459            -0.18663437
nodeifactor.orglevel.2 nodeifactor.orglevel.3 nodeifactor.orglevel.4
           0.65643869            0.43383710             0.22003495
nodeifactor.orglevel.5 nodeofactor.orglevel.2 nodeofactor.orglevel.3
           0.55165759           -0.97260773            -0.37348499
nodeofactor.orglevel.4 nodeofactor.orglevel.5       absdiff.orglevel
          -0.46248482           -0.02719759            -0.08542977
Starting unconstrained MCMC...
Sampling ==================>-----------   64% |  ETA:  2s
Back from unconstrained MCMC.
New interval = 512.
Estimated gradient of the log-likelihood:
                edges                mutual    gwesp.OTP.fixed.0.5
           -205.05600            -83.79495            -351.54695
     nodematch.gender      nodematch.region     nodematch.orglevel
           -131.20779            -97.68189             -38.15916
nodeifactor.orglevel.2 nodeifactor.orglevel.3 nodeifactor.orglevel.4
            -29.14295            -69.20547             -46.13937
nodeifactor.orglevel.5 nodeofactor.orglevel.2 nodeofactor.orglevel.3
            -35.41326            -23.39705             -70.04737
nodeofactor.orglevel.4 nodeofactor.orglevel.5       absdiff.orglevel
            -46.77221            -37.14947            -285.96989
Starting MCMLE Optimization...

Warning: 'glpk' selected as the solver, but package 'Rglpk' is not available;
falling back to 'lpSolveAPI'. This should be fine unless the sample size and/or
the number of parameters is very big.
```

```
1 Optimizing with step length 0.2727.
Using lognormal metric (see control.ergm function).
Using log-normal approx (no optim)
The log-likelihood improved by 5.0400.
Estimating equations are not within tolerance region.

Iteration 2 of at most 10 with free parameter vector:
                edges                mutual    gwesp.OTP.fixed.0.5
          -4.19508878            3.32210505             0.92504504
     nodematch.gender      nodematch.region     nodematch.orglevel
           0.01519900            0.02660961            -0.17118518
nodeifactor.orglevel.2 nodeifactor.orglevel.3 nodeifactor.orglevel.4
           0.72772571            0.41223661             0.29018863
nodeifactor.orglevel.5 nodeofactor.orglevel.2 nodeofactor.orglevel.3
           0.42082283           -0.88540850            -0.42008113
nodeofactor.orglevel.4 nodeofactor.orglevel.5       absdiff.orglevel
          -0.39754493           -0.18246573            -0.07251532
Starting unconstrained MCMC...
Back from unconstrained MCMC.
New interval = 256.
Estimated gradient of the log-likelihood:
                edges                mutual    gwesp.OTP.fixed.0.5
           123.535830             45.472588             161.515184
     nodematch.gender      nodematch.region     nodematch.orglevel
            76.158429             70.282646              25.889273
nodeifactor.orglevel.2 nodeifactor.orglevel.3 nodeifactor.orglevel.4
            23.869415             30.127887              50.803907
nodeifactor.orglevel.5 nodeofactor.orglevel.2 nodeofactor.orglevel.3
             7.771854             18.913123              29.320203
nodeofactor.orglevel.4 nodeofactor.orglevel.5       absdiff.orglevel
            52.661127              7.326570             157.113318
Starting MCMLE Optimization...
1 Optimizing with step length 0.6179.
Using lognormal metric (see control.ergm function).
Using log-normal approx (no optim)
The log-likelihood improved by 18.7610.
Distance from origin on tolerance region scale: 982.5427 (previously 3481.329).
Estimating equations are not within tolerance region.

Iteration 3 of at most 10 with free parameter vector:
                edges                mutual    gwesp.OTP.fixed.0.5
          -3.88680812            2.69599348             1.32254229
     nodematch.gender      nodematch.region     nodematch.orglevel
```

```
          -0.18428365             0.25496865            -0.60772743
nodeifactor.orglevel.2 nodeifactor.orglevel.3 nodeifactor.orglevel.4
           0.68341047             0.20236222             0.64112426
nodeifactor.orglevel.5 nodeofactor.orglevel.2 nodeofactor.orglevel.3
          -0.08550677            -0.94318561            -0.74770597
nodeofactor.orglevel.4 nodeofactor.orglevel.5       absdiff.orglevel
          -0.04998629            -0.88771806            -0.14633974
Starting unconstrained MCMC...
Back from unconstrained MCMC.
New interval = 128.
Estimated gradient of the log-likelihood:
                edges                  mutual    gwesp.OTP.fixed.0.5
            -422.22739             -163.02840             -771.29399
     nodematch.gender       nodematch.region     nodematch.orglevel
            -252.90445             -241.33812              -95.07310
nodeifactor.orglevel.2 nodeifactor.orglevel.3 nodeifactor.orglevel.4
             -81.55126              -75.69182             -202.29537
nodeifactor.orglevel.5 nodeofactor.orglevel.2 nodeofactor.orglevel.3
             -14.18280              -68.25190              -73.73344
nodeofactor.orglevel.4 nodeofactor.orglevel.5       absdiff.orglevel
            -208.78235              -14.51302             -552.35063
Starting MCMLE Optimization...
1 Optimizing with step length 0.1028.
Using lognormal metric (see control.ergm function).
Using log-normal approx (no optim)
The log-likelihood improved by 4.3234.
Distance from origin on tolerance region scale: 8188.224 (previously 8611.399).
Estimating equations are not within tolerance region.

Iteration 4 of at most 10 with free parameter vector:
                edges                  mutual    gwesp.OTP.fixed.0.5
          -4.276024672            2.805692623             1.382943785
     nodematch.gender       nodematch.region     nodematch.orglevel
          -0.147780682             0.225307327           -0.573968861
nodeifactor.orglevel.2 nodeifactor.orglevel.3 nodeifactor.orglevel.4
           0.708828742             0.247491146            0.621124438
nodeifactor.orglevel.5 nodeofactor.orglevel.2 nodeofactor.orglevel.3
          -0.002707387           -0.906438567           -0.704879319
nodeofactor.orglevel.4 nodeofactor.orglevel.5       absdiff.orglevel
          -0.093804813           -0.760212633           -0.129230301
Starting unconstrained MCMC...
Back from unconstrained MCMC.
New interval = 128.
```

```
Estimated gradient of the log-likelihood:
                edges                mutual    gwesp.OTP.fixed.0.5
            -374.43368            -145.04828            -683.43447
      nodematch.gender       nodematch.region     nodematch.orglevel
            -224.30919            -212.25600             -85.18667
nodeifactor.orglevel.2 nodeifactor.orglevel.3 nodeifactor.orglevel.4
             -73.21502             -65.26119            -180.55467
nodeifactor.orglevel.5 nodeofactor.orglevel.2 nodeofactor.orglevel.3
             -12.36786             -62.35144             -63.63130
nodeofactor.orglevel.4 nodeofactor.orglevel.5       absdiff.orglevel
            -185.38611             -12.87481            -492.94189
Starting MCMLE Optimization...
1 Optimizing with step length 0.1243.
Using lognormal metric (see control.ergm function).
Using log-normal approx (no optim)
The log-likelihood improved by 4.3680.
Distance from origin on tolerance region scale: 5654.691 (previously 7139.32).
Estimating equations are not within tolerance region.

Iteration 5 of at most 10 with free parameter vector:
                edges                mutual    gwesp.OTP.fixed.0.5
            -4.7360770             2.8466032             1.5596106
      nodematch.gender       nodematch.region     nodematch.orglevel
            -0.1248178             0.2035270            -0.5816492
nodeifactor.orglevel.2 nodeifactor.orglevel.3 nodeifactor.orglevel.4
             0.7102055             0.2738616             0.5746020
nodeifactor.orglevel.5 nodeofactor.orglevel.2 nodeofactor.orglevel.3
             0.1433184            -0.9140883            -0.6896789
nodeofactor.orglevel.4 nodeofactor.orglevel.5       absdiff.orglevel
            -0.1827212            -0.7032747            -0.1384523
Starting unconstrained MCMC...
Back from unconstrained MCMC.
New interval = 64.
Estimated gradient of the log-likelihood:
                edges                mutual    gwesp.OTP.fixed.0.5
            -321.26021            -124.12042            -587.38701
      nodematch.gender       nodematch.region     nodematch.orglevel
            -193.95242            -184.48737             -74.22758
nodeifactor.orglevel.2 nodeifactor.orglevel.3 nodeifactor.orglevel.4
             -62.43158             -54.60568            -157.53200
nodeifactor.orglevel.5 nodeofactor.orglevel.2 nodeofactor.orglevel.3
             -10.50884             -53.02611             -53.25137
nodeofactor.orglevel.4 nodeofactor.orglevel.5       absdiff.orglevel
```

```
           -160.88484              -10.78653             -419.11453
Starting MCMLE Optimization...
1 Optimizing with step length 0.1365.
Using lognormal metric (see control.ergm function).
Using log-normal approx (no optim)
The log-likelihood improved by 3.2358.
Distance from origin on tolerance region scale: 3473.638 (previously 4686.065).
Estimating equations are not within tolerance region.

Iteration 6 of at most 10 with free parameter vector:
                 edges                 mutual    gwesp.OTP.fixed.0.5
           -4.79666181             2.85835433             1.57463180
      nodematch.gender       nodematch.region     nodematch.orglevel
           -0.09268367             0.20675381            -0.63677013
nodeifactor.orglevel.2 nodeifactor.orglevel.3 nodeifactor.orglevel.4
            0.67044177             0.25525713             0.50662392
nodeifactor.orglevel.5 nodeofactor.orglevel.2 nodeofactor.orglevel.3
            0.17122602            -0.93274061            -0.66279091
nodeofactor.orglevel.4 nodeofactor.orglevel.5       absdiff.orglevel
           -0.26137751            -0.60783310            -0.17633296
Starting unconstrained MCMC...
Back from unconstrained MCMC.
New interval = 32.
Estimated gradient of the log-likelihood:
                 edges                 mutual    gwesp.OTP.fixed.0.5
           -229.809431             -89.222947            -426.686955
      nodematch.gender       nodematch.region     nodematch.orglevel
           -130.358540            -135.843329             -56.665363
nodeifactor.orglevel.2 nodeifactor.orglevel.3 nodeifactor.orglevel.4
            -39.054976             -38.584963            -116.070621
nodeifactor.orglevel.5 nodeofactor.orglevel.2 nodeofactor.orglevel.3
             -6.141026             -32.559322             -38.892873
nodeofactor.orglevel.4 nodeofactor.orglevel.5       absdiff.orglevel
           -116.922642              -6.799000            -290.452847
Starting MCMLE Optimization...
1 Optimizing with step length 0.2155.
Using lognormal metric (see control.ergm function).
Using log-normal approx (no optim)
The log-likelihood improved by 2.6648.
Distance from origin on tolerance region scale: 1147.799 (previously 1867.651).
Estimating equations are not within tolerance region.

Iteration 7 of at most 10 with free parameter vector:
```

```
                edges                mutual    gwesp.OTP.fixed.0.5
          -4.64138120            2.89398647            1.37878156
      nodematch.gender      nodematch.region     nodematch.orglevel
          -0.05727926            0.16039711            -0.66137175
nodeifactor.orglevel.2 nodeifactor.orglevel.3 nodeifactor.orglevel.4
           0.74468584            0.32174322             0.54975080
nodeifactor.orglevel.5 nodeofactor.orglevel.2 nodeofactor.orglevel.3
           0.33664394           -0.88925520            -0.56006893
nodeofactor.orglevel.4 nodeofactor.orglevel.5       absdiff.orglevel
          -0.22484101           -0.61441111            -0.16972965
Starting unconstrained MCMC...
Back from unconstrained MCMC.
New interval = 16.
Estimated gradient of the log-likelihood:
                edges                mutual    gwesp.OTP.fixed.0.5
            152.9916              53.9768              177.4393
      nodematch.gender      nodematch.region     nodematch.orglevel
             94.5440              84.6900               32.2876
nodeifactor.orglevel.2 nodeifactor.orglevel.3 nodeifactor.orglevel.4
             29.9312              38.6288               59.4528
nodeifactor.orglevel.5 nodeofactor.orglevel.2 nodeofactor.orglevel.3
             11.0104              23.0648               37.5444
nodeofactor.orglevel.4 nodeofactor.orglevel.5       absdiff.orglevel
             61.8216              11.0800              196.2360
Starting MCMLE Optimization...
1 Optimizing with step length 0.0402.
Using lognormal metric (see control.ergm function).
Using log-normal approx (no optim)
The log-likelihood improved by 10.0716.
Distance from origin on tolerance region scale: 124434.3 (previously 853626.9).
Estimating equations are not within tolerance region.

Iteration 8 of at most 10 with free parameter vector:
                edges                mutual    gwesp.OTP.fixed.0.5
          -4.058974484           1.629090534            4.726287704
      nodematch.gender      nodematch.region     nodematch.orglevel
          -0.200966928          -0.005805028           -0.568803852
nodeifactor.orglevel.2 nodeifactor.orglevel.3 nodeifactor.orglevel.4
           0.404898666          -0.036348987            0.150726398
nodeifactor.orglevel.5 nodeofactor.orglevel.2 nodeofactor.orglevel.3
          -0.218022393          -0.842278281           -0.814746709
nodeofactor.orglevel.4 nodeofactor.orglevel.5       absdiff.orglevel
          -0.453916694          -1.084539731           -0.208980100
```

```
Starting unconstrained MCMC...
Back from unconstrained MCMC.


Warning: The following linear dependence has been detected among the model statistics:
   16568/10049 * edges = gwesp.OTP.fixed.0.5
This may indicate that the model is nonidentifiable.


New interval = 16.
Estimated gradient of the log-likelihood:
                 edges                 mutual    gwesp.OTP.fixed.0.5
            -1447.7663              -741.4593             -2485.8265
      nodematch.gender       nodematch.region     nodematch.orglevel
             -899.7311              -718.1320              -340.8545
nodeifactor.orglevel.2 nodeifactor.orglevel.3 nodeifactor.orglevel.4
             -282.7084              -351.8198              -526.4104
nodeifactor.orglevel.5 nodeofactor.orglevel.2 nodeofactor.orglevel.3
             -105.4881              -289.1289              -352.2113
nodeofactor.orglevel.4 nodeofactor.orglevel.5       absdiff.orglevel
             -524.4153              -105.7352             -1895.2179
Starting MCMLE Optimization...
1 Optimizing with step length 0.0043.
Using lognormal metric (see control.ergm function).
Using log-normal approx (no optim)
The log-likelihood improved by 6.9733.
Distance from origin on tolerance region scale: Inf (previously Inf).
Estimating equations are not within tolerance region.

Iteration 9 of at most 10 with free parameter vector:
                 edges                 mutual    gwesp.OTP.fixed.0.5
           -6.81099832             6.04582502             3.05710087
      nodematch.gender       nodematch.region     nodematch.orglevel
            0.13560186            -0.04149581             0.01293985
nodeifactor.orglevel.2 nodeifactor.orglevel.3 nodeifactor.orglevel.4
            0.85082403             0.33954256             0.31673411
nodeifactor.orglevel.5 nodeofactor.orglevel.2 nodeofactor.orglevel.3
           -0.28218989            -0.83497736            -0.38620429
nodeofactor.orglevel.4 nodeofactor.orglevel.5       absdiff.orglevel
           -0.32912555            -0.66943684             0.06675152
Starting unconstrained MCMC...
Sampling ======================>-------   78% |  ETA:  1s
Back from unconstrained MCMC.
```

```
Warning: The following linear dependence has been detected among the model statistics:
   16568/10049 * edges = gwesp.OTP.fixed.0.5
This may indicate that the model is nonidentifiable.


New interval = 64.
Estimated gradient of the log-likelihood:
                  edges                  mutual    gwesp.OTP.fixed.0.5
           -1338.04674             -687.76728            -2304.92952
      nodematch.gender       nodematch.region     nodematch.orglevel
            -839.46002             -662.55456             -316.25560
nodeifactor.orglevel.2 nodeifactor.orglevel.3 nodeifactor.orglevel.4
            -263.02762             -326.67149             -490.01284
nodeifactor.orglevel.5 nodeofactor.orglevel.2 nodeofactor.orglevel.3
             -87.93810             -268.29683             -327.99334
nodeofactor.orglevel.4 nodeofactor.orglevel.5       absdiff.orglevel
            -487.77289              -88.72798            -1753.37457
Starting MCMLE Optimization...
1 Optimizing with step length 0.0230.
Using lognormal metric (see control.ergm function).
Using log-normal approx (no optim)
The log-likelihood improved by 2.8818.
Distance from origin on tolerance region scale: Inf (previously Inf).
Estimating equations are not within tolerance region.
Estimating equations did not move closer to tolerance region more than 1 time(s) in 4 steps;

Iteration 10 of at most 10 with free parameter vector:
                  edges                  mutual    gwesp.OTP.fixed.0.5
           -6.99507725             6.23511915             2.94545136
      nodematch.gender       nodematch.region     nodematch.orglevel
            0.14417513            -0.04369960             0.04682927
nodeifactor.orglevel.2 nodeifactor.orglevel.3 nodeifactor.orglevel.4
            0.90936060             0.35746618             0.29572080
nodeifactor.orglevel.5 nodeofactor.orglevel.2 nodeofactor.orglevel.3
           -0.11588062            -0.81005121            -0.36341598
nodeofactor.orglevel.4 nodeofactor.orglevel.5       absdiff.orglevel
           -0.23917422            -0.50835436             0.07167377
Starting unconstrained MCMC...
Sampling ==========>--------------------   32% |  ETA:  2s
Back from unconstrained MCMC.


Warning: The following linear dependence has been detected among the model statistics:
   16568/10049 * edges = gwesp.OTP.fixed.0.5
```

```
This may indicate that the model is nonidentifiable.


New interval = 128.
Estimated gradient of the log-likelihood:
                  edges                  mutual    gwesp.OTP.fixed.0.5
           -1301.23916             -668.76084            -2244.24408
      nodematch.gender       nodematch.region     nodematch.orglevel
            -817.77932             -643.71772             -308.88430
nodeifactor.orglevel.2 nodeifactor.orglevel.3 nodeifactor.orglevel.4
            -256.86262             -314.94517             -477.33736
nodeifactor.orglevel.5 nodeofactor.orglevel.2 nodeofactor.orglevel.3
             -88.20749             -261.66880             -316.06481
nodeofactor.orglevel.4 nodeofactor.orglevel.5       absdiff.orglevel
            -475.76688              -88.87161            -1705.77563
Starting MCMLE Optimization...
1 Optimizing with step length 0.0315.
Using lognormal metric (see control.ergm function).
Using log-normal approx (no optim)
Starting MCMC s.e. computation.
The log-likelihood improved by 534566296490.3445.
Distance from origin on tolerance region scale: Inf (previously Inf).
Estimating equations are not within tolerance region.
MCMLE estimation did not converge after 10 iterations. The estimated coefficients may not be
Finished MCMLE.
Evaluating log-likelihood at the estimate. Initializing model to obtain the list of dyad-
independent terms...
Fitting the dyad-independent submodel...
Dyad-independent submodel MLE has likelihood -536.9021 at:
 [1] -2.15472501  0.00000000  0.00000000 -0.08148573  0.22428405 -0.47643262
 [7]  0.29200793  0.26881856  0.41095461  0.26340708 -0.42568630 -0.15863497
[13]  0.05487618 -0.13535062 -0.13366608  0.00000000
Bridging between the dyad-independent submodel and the full model...
Setting up bridge sampling...
Initializing model and proposals...
Model and proposals initialized.
Using 16 bridges: Running theta=[ 5.474717e+11,-2.170970e+02,-3.320584e+11,-
1.890251e+01,-2.247607e+02,-2.550273e+02, 1.394022e+02, 2.398214e+01,-4.793101e+01, 1.293805e
1.093491e+02,-9.364157e+01,-2.722199e+02, 5.754192e+02,-5.918668e+01, 0.000000e+00].
Running theta=[ 5.121509e+11,-2.030907e+02,-3.106353e+11,-1.768825e+01,-
2.102455e+02,-2.386047e+02, 1.304273e+02, 2.245225e+01,-4.481217e+01, 1.210504e+02,-
1.023218e+02,-8.761041e+01,-2.546537e+02, 5.382866e+02,-5.537681e+01, 0.000000e+00].
Running theta=[ 4.768302e+11,-1.890845e+02,-2.892121e+11,-1.647399e+01,-
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1.957304e+02,-2.221821e+02, 1.214525e+02, 2.092236e+01,-4.169334e+01, 1.127203e+02,-
9.529450e+01,-8.157926e+01,-2.370876e+02, 5.011541e+02,-5.156694e+01, 0.000000e+00].
Running theta=[ 4.415094e+11,-1.750782e+02,-2.677890e+11,-1.525973e+01,-
1.812152e+02,-2.057594e+02, 1.124776e+02, 1.939247e+01,-3.857450e+01, 1.043901e+02,-
8.826718e+01,-7.554810e+01,-2.195215e+02, 4.640215e+02,-4.775706e+01, 0.000000e+00].
Running theta=[ 4.061887e+11,-1.610720e+02,-2.463659e+11,-1.404547e+01,-
1.667001e+02,-1.893368e+02, 1.035028e+02, 1.786257e+01,-3.545566e+01, 9.605998e+01,-
8.123986e+01,-6.951694e+01,-2.019554e+02, 4.268890e+02,-4.394719e+01, 0.000000e+00].
Running theta=[ 3.708679e+11,-1.470657e+02,-2.249428e+11,-1.283121e+01,-
1.521849e+02,-1.729141e+02, 9.452792e+01, 1.633268e+01,-3.233683e+01, 8.772984e+01,-
7.421254e+01,-6.348578e+01,-1.843893e+02, 3.897564e+02,-4.013732e+01, 0.000000e+00].
Running theta=[ 3.355472e+11,-1.330594e+02,-2.035196e+11,-1.161695e+01,-
1.376697e+02,-1.564915e+02, 8.555307e+01, 1.480279e+01,-2.921799e+01, 7.939971e+01,-
6.718522e+01,-5.745463e+01,-1.668232e+02, 3.526239e+02,-3.632745e+01, 0.000000e+00].
Running theta=[ 3.002264e+11,-1.190532e+02,-1.820965e+11,-1.040269e+01,-
1.231546e+02,-1.400689e+02, 7.657822e+01, 1.327290e+01,-2.609916e+01, 7.106957e+01,-
6.015790e+01,-5.142347e+01,-1.492571e+02, 3.154913e+02,-3.251758e+01, 0.000000e+00].
Running theta=[ 2.649057e+11,-1.050469e+02,-1.606734e+11,-9.188435e+00,-
1.086394e+02,-1.236462e+02, 6.760337e+01, 1.174301e+01,-2.298032e+01, 6.273943e+01,-
5.313058e+01,-4.539231e+01,-1.316910e+02, 2.783588e+02,-2.870771e+01, 0.000000e+00].
Running theta=[ 2.295849e+11,-9.104067e+01,-1.392503e+11,-7.974175e+00,-
9.412426e+01,-1.072236e+02, 5.862852e+01, 1.021312e+01,-1.986148e+01, 5.440930e+01,-
4.610326e+01,-3.936116e+01,-1.141248e+02, 2.412262e+02,-2.489783e+01, 0.000000e+00].
Running theta=[ 1.942642e+11,-7.703442e+01,-1.178272e+11,-6.759915e+00,-
7.960910e+01,-9.080095e+01, 4.965367e+01, 8.683224e+00,-1.674265e+01, 4.607916e+01,-
3.907594e+01,-3.333000e+01,-9.655874e+01, 2.040937e+02,-2.108796e+01, 0.000000e+00].
Running theta=[ 1.589434e+11,-6.302816e+01,-9.640404e+10,-5.545655e+00,-
6.509394e+01,-7.437831e+01, 4.067883e+01, 7.153332e+00,-1.362381e+01, 3.774902e+01,-
3.204862e+01,-2.729884e+01,-7.899263e+01, 1.669611e+02,-1.727809e+01, 0.000000e+00].
Running theta=[ 1.236226e+11,-4.902190e+01,-7.498092e+10,-4.331395e+00,-
5.057878e+01,-5.795567e+01, 3.170398e+01, 5.623440e+00,-1.050497e+01, 2.941889e+01,-
2.502130e+01,-2.126768e+01,-6.142651e+01, 1.298286e+02,-1.346822e+01, 0.000000e+00].
Running theta=[ 8.830189e+10,-3.501564e+01,-5.355780e+10,-3.117135e+00,-
3.606362e+01,-4.153303e+01, 2.272913e+01, 4.093548e+00,-7.386137e+00, 2.108875e+01,-
1.799398e+01,-1.523653e+01,-4.386040e+01, 9.269602e+01,-9.658346e+00, 0.000000e+00].
Running theta=[ 5.298113e+10,-2.100939e+01,-3.213468e+10,-1.902876e+00,-
2.154846e+01,-2.511039e+01, 1.375428e+01, 2.563656e+00,-4.267300e+00, 1.275861e+01,-
1.096667e+01,-9.205371e+00,-2.629429e+01, 5.556347e+01,-5.848474e+00, 0.000000e+00].
Running theta=[ 1.766038e+10,-7.003129e+00,-1.071156e+10,-6.886157e-01,-
7.033296e+00,-8.687752e+00, 4.779432e+00, 1.033764e+00,-1.148464e+00, 4.428476e+00,-
3.939346e+00,-3.174214e+00,-8.728180e+00, 1.843092e+01,-2.038602e+00, 0.000000e+00].
.
Bridge sampling finished. Collating...
```

```
Bridging finished.

This model was fit using MCMC.  To examine model diagnostics and check
for degeneracy, use the mcmc.diagnostics() function.
```

```
summary(ergm.model)
```

```
Call:
ergm(formula = consult ~ edges + mutual + gwesp(0.5, fixed = T) +
    nodematch("gender") + nodematch("region") + nodematch("orglevel") +
    nodeifactor("orglevel") + nodeofactor("orglevel") + absdiff("orglevel"),
    control = control.ergm(MCMC.samplesize = 5000, MCMC.burnin = 10000,
        MCMLE.maxit = 10), verbose = T)

Monte Carlo Maximum Likelihood Results:

                         Estimate Std. Error MCMC %    z value Pr(>|z|)
edges                   5.651e+11  1.981e-02    100  2.852e+13   <1e-04 ***
mutual                 -2.241e+02  1.446e-02    100 -1.549e+04   <1e-04 ***
gwesp.OTP.fixed.0.5    -3.428e+11  1.202e-02    100 -2.852e+13   <1e-04 ***
nodematch.gender       -1.951e+01  1.351e-02    100 -1.444e+03   <1e-04 ***
nodematch.region       -2.320e+02  1.298e-02    100 -1.787e+04   <1e-04 ***
nodematch.orglevel     -2.632e+02  2.277e-02    100 -1.156e+04   <1e-04 ***
nodeifactor.orglevel.2  1.439e+02  2.534e-02    100  5.677e+03   <1e-04 ***
nodeifactor.orglevel.3  2.475e+01  2.542e-02    100  9.737e+02   <1e-04 ***
nodeifactor.orglevel.4 -4.949e+01  2.373e-02    100 -2.085e+03   <1e-04 ***
nodeifactor.orglevel.5  1.335e+02  2.980e-02    100  4.482e+03   <1e-04 ***
nodeofactor.orglevel.2 -1.129e+02  2.456e-02    100 -4.595e+03   <1e-04 ***
nodeofactor.orglevel.3 -9.666e+01  2.417e-02    100 -4.000e+03   <1e-04 ***
nodeofactor.orglevel.4 -2.810e+02  2.222e-02    100 -1.265e+04   <1e-04 ***
nodeofactor.orglevel.5  5.940e+02  2.745e-02    100  2.164e+04   <1e-04 ***
absdiff.orglevel       -6.109e+01  1.038e-02    100 -5.886e+03   <1e-04 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

     Null Deviance: 2.274e+03  on 1640  degrees of freedom
 Residual Deviance: 4.479e+14  on 1625  degrees of freedom

AIC: 4.479e+14  BIC: 4.479e+14  (Smaller is better. MC Std. Err. = 5.98e+11)
Warnings:

 * The following linear dependence has been detected among the model
```

```
   statistics and/or estimating functions:

   16568/10049 * edges = gwesp.OTP.fixed.0.5
```

Here are the results provided for the questions:

```
Monte Carlo Maximum Likelihood Results:

                         Estimate Std. Error MCMC %    z value Pr(>|z|)
edges                   -4.064e+12  4.939e-01      0 -8.229e+12  < 1e-04 ***
mutual                   4.247e+00  5.186e-01      0  8.189e+00  < 1e-04 ***
gwesp.OTP.fixed.0.5      2.465e+12  2.995e-01      0  8.229e+12  < 1e-04 ***
nodematch.gender         1.404e-01  2.384e-01      1  5.890e-01 0.555892
nodematch.region        -9.278e-01  2.830e-01      1 -3.278e+00 0.001045 **
nodematch.orglevel       1.050e+00  4.258e-01      1  2.467e+00 0.013635 *
nodeifactor.orglevel.2   6.605e-01  8.150e-01      0  8.100e-01 0.417693
nodeifactor.orglevel.3   9.048e-01  8.049e-01      0  1.124e+00 0.260950
nodeifactor.orglevel.4   1.721e+00  7.317e-01      0  2.352e+00 0.018663 *
nodeifactor.orglevel.5   8.214e-01  1.077e+00      0  7.620e-01 0.445845
nodeofactor.orglevel.2   1.688e+00  8.093e-01      0  2.086e+00 0.037010 *
nodeofactor.orglevel.3   3.023e+00  8.392e-01      0  3.602e+00 0.000315 ***
nodeofactor.orglevel.4   2.141e+00  7.792e-01      0  2.748e+00 0.005992 **
nodeofactor.orglevel.5  -1.358e+00  1.139e+00      0 -1.193e+00 0.232937
absdiff.orglevel         5.002e-01  1.642e-01      1  3.045e+00 0.002323 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

     Null Deviance: 2.274e+03  on 1640  degrees of freedom
 Residual Deviance: 4.548e+14  on 1625  degrees of freedom
```

Using the results provided, here are the answers to the questions:

```
•1: Based on visual inspection only, there seems to be a regional homophily between nodes, me
•2: The Monte Carlo Maximum Likelihood results are not significant for gender matching, meani
•3: The Monte Carlo Maximum Likelihood results are significant but negative for region matchi
•4: It is difficult to visually guess homophily or heterophily at an organizational level. Th
•5: Reading nodeifactor tests, vertices at the 4th organizational level seem to moderately at
•6: The nodeofactor() terms measures the effect of factor attributes on outward edges. In mor
•7: The absdiff() term measures the homophily based on continuous attributes instead of binar
```

On your own

Running the ERGM model on my own led to an error message (Error in ergm.MCMLE(init, s, s.obs, control = control, verbose = verbose, : Unconstrained MCMC sampling did not mix at all. Optimization cannot continue.). Writing the summary despite the message error shows unconvincing results:

```
# Monte Carlo Maximum Likelihood Results:

#   Estimate Std. Error MCMC %    z value Pr(>|z|)
#  edges                   5.651e+11  1.981e-02    100  2.852e+13   <1e-04 ***
#  mutual                 -2.241e+02  1.446e-02    100 -1.549e+04   <1e-04 ***
#  gwesp.OTP.fixed.0.5    -3.428e+11  1.202e-02    100 -2.852e+13   <1e-04 ***
#  nodematch.gender       -1.951e+01  1.351e-02    100 -1.444e+03   <1e-04 ***
#  nodematch.region       -2.320e+02  1.298e-02    100 -1.787e+04   <1e-04 ***
#  nodematch.orglevel     -2.632e+02  2.277e-02    100 -1.156e+04   <1e-04 ***
#  nodeifactor.orglevel.2  1.439e+02  2.534e-02    100  5.677e+03   <1e-04 ***
#  nodeifactor.orglevel.3  2.475e+01  2.542e-02    100  9.737e+02   <1e-04 ***
#  nodeifactor.orglevel.4 -4.949e+01  2.373e-02    100 -2.085e+03   <1e-04 ***
#  nodeifactor.orglevel.5  1.335e+02  2.980e-02    100  4.482e+03   <1e-04 ***
#  nodeofactor.orglevel.2 -1.129e+02  2.456e-02    100 -4.595e+03   <1e-04 ***
#  nodeofactor.orglevel.3 -9.666e+01  2.417e-02    100 -4.000e+03   <1e-04 ***
#  nodeofactor.orglevel.4 -2.810e+02  2.222e-02    100 -1.265e+04   <1e-04 ***
#  nodeofactor.orglevel.5  5.940e+02  2.745e-02    100  2.164e+04   <1e-04 ***
#  absdiff.orglevel       -6.109e+01  1.038e-02    100 -5.886e+03   <1e-04 ***
#  ---
#  Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#
# Null Deviance: 2.274e+03  on 1640  degrees of freedom
# Residual Deviance: 4.479e+14  on 1625  degrees of freedom
#
# AIC: 4.479e+14  BIC: 4.479e+14  (Smaller is better. MC Std. Err. = 5.98e+11)
# Warnings:
#
#  * The following linear dependence has been detected among the model statistics and/or esti
#
#  16568/10049 * edges = gwesp.OTP.fixed.0.5
```

Using a simpler model, removing gwesp specification showed new & different results:

```
set.seed(510)
ergm.model <- ergm(consult ~ edges + mutual +
```

```
                      nodematch("gender") + nodematch("region") +
                      nodematch("orglevel") + nodeifactor("orglevel") +
                      nodeofactor("orglevel") + absdiff("orglevel"), control = control.ergm(
                        MCMC.samplesize = 5000,
                        MCMC.burnin = 10000,
                        MCMLE.maxit = 10
                      ),
                    verbose = T)
```

```
Evaluating network in model.


Initializing unconstrained Metropolis-Hastings proposal: 'ergm:MH_SPDyad'.
Initializing model...
Model initialized.
Using initial method 'MPLE'.
Initial parameters provided by caller: None.
  number of free parameters: 14
  number of fixed parameters: 0
Fitting initial model.
Starting maximum pseudolikelihood estimation (MPLE):
Obtaining the responsible dyads.
Evaluating the predictor and response matrix.
Maximizing the pseudolikelihood.
Finished MPLE.
Starting Monte Carlo maximum likelihood estimation (MCMLE):
Density guard set to 10000 from an initial count of 168 edges.

Iteration 1 of at most 10 with free parameter vector:
                  edges                  mutual       nodematch.gender
            -3.35790556             3.96650700            -0.02411897
      nodematch.region     nodematch.orglevel nodeifactor.orglevel.2
             0.14212366            -0.32042434             0.91533749
nodeifactor.orglevel.3 nodeifactor.orglevel.4 nodeifactor.orglevel.5
             0.61636765             0.59148024             0.81681092
nodeofactor.orglevel.2 nodeofactor.orglevel.3 nodeofactor.orglevel.4
            -0.93136061            -0.38813142            -0.26536741
nodeofactor.orglevel.5       absdiff.orglevel
            -0.16190383            -0.08948597
Starting unconstrained MCMC...
Back from unconstrained MCMC.
New interval = 512.
```

```
Estimated gradient of the log-likelihood:
                edges                mutual       nodematch.gender
           -1.7985263            -1.2370526            -2.5172632
      nodematch.region     nodematch.orglevel nodeifactor.orglevel.2
           -0.6383158             2.3406316             0.6705263
nodeifactor.orglevel.3 nodeifactor.orglevel.4 nodeifactor.orglevel.5
           -1.5004211             2.4915789            -4.5031579
nodeofactor.orglevel.2 nodeofactor.orglevel.3 nodeofactor.orglevel.4
            0.9768421            -2.3183158             2.5370526
nodeofactor.orglevel.5       absdiff.orglevel
           -4.5016842            -4.6709474
Starting MCMLE Optimization...
1 Optimizing with step length 1.0000.
Using lognormal metric (see control.ergm function).
Using log-normal approx (no optim)
The log-likelihood improved by 1.2723.
Estimating equations are not within tolerance region.

Iteration 2 of at most 10 with free parameter vector:
                edges                mutual       nodematch.gender
          -3.27655171            3.94587903           -0.05878478
      nodematch.region     nodematch.orglevel nodeifactor.orglevel.2
           0.14160548            -0.25555766             0.89120237
nodeifactor.orglevel.3 nodeifactor.orglevel.4 nodeifactor.orglevel.5
           0.58636529             0.60323445             0.57594085
nodeofactor.orglevel.2 nodeofactor.orglevel.3 nodeofactor.orglevel.4
          -0.92195988            -0.49578885           -0.28514867
nodeofactor.orglevel.5       absdiff.orglevel
          -0.41602402            -0.07465258
Starting unconstrained MCMC...
Back from unconstrained MCMC.
New interval = 256.
Estimated gradient of the log-likelihood:
                edges                mutual       nodematch.gender
          -2.37957895            -1.03326316           -1.29684211
      nodematch.region     nodematch.orglevel nodeifactor.orglevel.2
          -1.31157895            -1.16378947           -1.29831579
nodeifactor.orglevel.3 nodeifactor.orglevel.4 nodeifactor.orglevel.5
           0.21010526            -0.63410526           -0.44905263
nodeofactor.orglevel.2 nodeofactor.orglevel.3 nodeofactor.orglevel.4
          -1.41410526             0.04084211           -0.38673684
nodeofactor.orglevel.5       absdiff.orglevel
          -0.55368421            -2.31221053
```

```
Starting MCMLE Optimization...
1 Optimizing with step length 1.0000.
Using lognormal metric (see control.ergm function).
Using log-normal approx (no optim)
Starting MCMC s.e. computation.
The log-likelihood improved by 0.0712.
Distance from origin on tolerance region scale: 1.423683 (previously 28.77277).
Test statistic: T^2 = 32.83715, with 14 free parameter(s) and 993.6671 degrees of freedom.
Convergence test p-value: 0.0039. Converged with 99% confidence.
Finished MCMLE.
Evaluating log-likelihood at the estimate. Initializing model to obtain the list of dyad-
independent terms...
Fitting the dyad-independent submodel...
Dyad-independent submodel MLE has likelihood -536.9021 at:
 [1] -2.15472501  0.00000000 -0.08148573  0.22428405 -0.47643262  0.29200793
 [7]  0.26881856  0.41095461  0.26340708 -0.42568630 -0.15863497  0.05487618
[13] -0.13535062 -0.13366608  0.00000000
Bridging between the dyad-independent submodel and the full model...
Setting up bridge sampling...
Initializing model and proposals...
Model and proposals initialized.
Using 16 bridges: Running theta=[-3.24635235, 3.82127537,-0.06140583, 0.14146883,-
0.28639261, 0.87032001, 0.60603158, 0.61136175, 0.56326227,-0.95785406,-
0.49001688,-0.27088985,-0.44896709,-0.07459542, 0.00000000].
Running theta=[-3.17592478, 3.57474148,-0.06270130, 0.14681175,-0.29865326, 0.83300955, 0.584
0.92352066,-0.46863740,-0.24987269,-0.42873377,-0.07840643, 0.00000000].
Running theta=[-3.10549721, 3.32820758,-0.06399678, 0.15215467,-0.31091390, 0.79569910, 0.562
0.88918725,-0.44725792,-0.22885552,-0.40850045,-0.08221744, 0.00000000].
Running theta=[-3.03506964, 3.08167369,-0.06529226, 0.15749759,-0.32317455, 0.75838864, 0.540
0.85485385,-0.42587844,-0.20783836,-0.38826713,-0.08602845, 0.00000000].
Running theta=[-2.96464207, 2.83513979,-0.06658774, 0.16284050,-0.33543520, 0.72107818, 0.519
0.82052044,-0.40449897,-0.18682120,-0.36803381,-0.08983946, 0.00000000].
Running theta=[-2.89421450, 2.58860590,-0.06788321, 0.16818342,-0.34769584, 0.68376773, 0.497
0.78618704,-0.38311949,-0.16580404,-0.34780049,-0.09365047, 0.00000000].
Running theta=[-2.82378693, 2.34207200,-0.06917869, 0.17352634,-0.35995649, 0.64645727, 0.475
0.75185364,-0.36174001,-0.14478687,-0.32756717,-0.09746148, 0.00000000].
Running theta=[-2.75335936, 2.09553811,-0.07047417, 0.17886925,-0.37221713, 0.60914681, 0.453
0.71752023,-0.34036053,-0.12376971,-0.30733385,-0.10127249, 0.00000000].
Running theta=[-2.68293179, 1.84900421,-0.07176965, 0.18421217,-0.38447778, 0.57183636, 0.431
0.68318683,-0.31898105,-0.10275255,-0.28710053,-0.10508350, 0.00000000].
Running theta=[-2.61250422, 1.60247032,-0.07306513, 0.18955509,-0.39673843, 0.53452590, 0.410
0.64885342,-0.29760158,-0.08173538,-0.26686720,-0.10889451, 0.00000000].
Running theta=[-2.54207665, 1.35593642,-0.07436060, 0.19489801,-0.40899907, 0.49721544, 0.388
```

```
0.61452002,-0.27622210,-0.06071822,-0.24663388,-0.11270552, 0.00000000].
Running theta=[-2.47164908, 1.10940253,-0.07565608, 0.20024092,-0.42125972, 0.45990499, 0.366
0.58018662,-0.25484262,-0.03970106,-0.22640056,-0.11651653, 0.00000000].
Running theta=[-2.40122151, 0.86286863,-0.07695156, 0.20558384,-0.43352036, 0.42259453, 0.344
0.54585321,-0.23346314,-0.01868389,-0.20616724,-0.12032754, 0.00000000].
Running theta=[-2.330793937, 0.616334737,-0.078247037, 0.210926756,-0.445781009, 0.385284073
0.511519808,-0.212083664, 0.002333269,-0.185933919,-0.124138552, 0.000000000].
Running theta=[-2.26036637, 0.36980084,-0.07954251, 0.21626967,-0.45804166, 0.34797362, 0.301
0.47718640,-0.19070419, 0.02335043,-0.16570060,-0.12794956, 0.00000000].
Running theta=[-2.18993880, 0.12326695,-0.08083799, 0.22161259,-0.47030230, 0.31066316, 0.279
0.44285300,-0.16932471, 0.04436759,-0.14546728,-0.13176057, 0.00000000].
.
Bridge sampling finished. Collating...
Bridging finished.

This model was fit using MCMC.  To examine model diagnostics and check
for degeneracy, use the mcmc.diagnostics() function.
```

```
summary(ergm.model)
```

```
Call:
ergm(formula = consult ~ edges + mutual + nodematch("gender") +
    nodematch("region") + nodematch("orglevel") + nodeifactor("orglevel") +
    nodeofactor("orglevel") + absdiff("orglevel"), control = control.ergm(MCMC.samplesize = 5
    MCMC.burnin = 10000, MCMLE.maxit = 10), verbose = T)

Monte Carlo Maximum Likelihood Results:

                      Estimate Std. Error MCMC % z value Pr(>|z|)
edges                 -3.28157    0.45510      0  -7.211   <1e-04 ***
mutual                 3.94454    0.31121      0  12.675   <1e-04 ***
nodematch.gender      -0.06076    0.14093      0  -0.431   0.6664
nodematch.region       0.13880    0.12556      0   1.105   0.2690
nodematch.orglevel    -0.28026    0.23816      0  -1.177   0.2393
nodeifactor.orglevel.2  0.88898    0.41665      0   2.134   0.0329 *
nodeifactor.orglevel.3  0.61691    0.40831      0   1.511   0.1308
nodeifactor.orglevel.4  0.61783    0.38768      0   1.594   0.1110
nodeifactor.orglevel.5  0.57294    0.51548      0   1.111   0.2664
nodeofactor.orglevel.2 -0.97502    0.39489      0  -2.469   0.0135 *
nodeofactor.orglevel.3 -0.50071    0.37262      0  -1.344   0.1790
nodeofactor.orglevel.4 -0.28140    0.34811      0  -0.808   0.4189
nodeofactor.orglevel.5 -0.45908    0.47931      0  -0.958   0.3382
```

```
absdiff.orglevel       -0.07269    0.10830      0  -0.671   0.5021
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

     Null Deviance: 2273.5  on 1640  degrees of freedom
 Residual Deviance:  883.9  on 1626  degrees of freedom

AIC: 911.9  BIC: 987.5  (Smaller is better. MC Std. Err. = 0.6663)
```

```
Monte Carlo Maximum Likelihood Results:

                      Estimate Std. Error MCMC % z value Pr(>|z|)
edges                 -3.27777    0.47170      0  -6.949   <1e-04 ***
mutual                 3.94090    0.31673      0  12.442   <1e-04 ***
nodematch.gender      -0.05306    0.13973      0  -0.380   0.7042
nodematch.region       0.13823    0.12806      0   1.079   0.2804
nodematch.orglevel    -0.28713    0.24681      0  -1.163   0.2447
nodeifactor.orglevel.2  0.87747    0.42041      0   2.087   0.0369 *
nodeifactor.orglevel.3  0.60088    0.40895      0   1.469   0.1417
nodeifactor.orglevel.4  0.60948    0.37956      0   1.606   0.1083
nodeifactor.orglevel.5  0.55657    0.51831      0   1.074   0.2829
nodeofactor.orglevel.2 -0.94591    0.38738      0  -2.442   0.0146 *
nodeofactor.orglevel.3 -0.47655    0.37009      0  -1.288   0.1979
nodeofactor.orglevel.4 -0.27701    0.34061      0  -0.813   0.4161
nodeofactor.orglevel.5 -0.44775    0.47656      0  -0.940   0.3474
absdiff.orglevel       -0.07782    0.11385      0  -0.684   0.4943
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

     Null Deviance: 2273.5  on 1640  degrees of freedom
 Residual Deviance:  880.1  on 1626  degrees of freedom

AIC: 908.1  BIC: 983.8  (Smaller is better. MC Std. Err. = 0.6431)
```

```
These new results contredict previous analysis: only one organizational level significantly a
```